Differential Geometry Excercise 2

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Question 1

Let $\psi : M \to N$ be a diffeomorphism and X, Y vector fields on M. Then, for each $f \in C^\infty(N)$:

$$d\psi X(f) = X(f \circ \psi) \qquad d\psi Y(f) = Y(f \circ \psi) \tag{1}$$

Thus, for every point $p \in N$ for which $q = \psi^{-1}(p) \in M$, we get

$$[d\psi X, d\psi Y](f)(p) = X(Y(f \circ \psi))(q) - Y(X(f \circ \psi))(q) = [X, Y](f \circ \psi)(q) = d\psi[X, Y](f)(p) \tag{2}$$

Question 2

For every three vector fields X, Y, and Z:

$$\begin{gathered}
[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = \\
= X\,[Y, Z] - [Y, Z]\,X + Y\,[Z, X] - [Z, X]\,Y + Z\,[X, Y] - [X, Y]\,Z = \\
= XYZ - XZY - YZX + ZYX+ \\
+YZX - YXZ - ZXY + XZY+ \\
+ZXY - ZYX - XYZ + YXZ = 0
\end{gathered} \tag{3}$$

Question 3

part 1

Let X be a field vector over a smooth manifold M such that $Xf = 0$ for any smooth scalar function $f \in C^\infty(M)$. We can write X using its local expression in a point $x \in M$:

$$X(x)(f) = \sum_i y_i(x) \frac{\partial f}{\partial x_i}(x) \tag{4}$$

We can now choose functions $\{f_i\}$ such that in the neighborhood of x:

$$\frac{\partial f_i}{\partial x_j}(x) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \tag{5}$$

These are smooth functions, so according to the assumption $Xf_i = 0$. Therefore, for each f_i, the LHS of equation 4 is zero. The RHS gives:

$$\sum_j y_j(x) \frac{\partial f_i}{\partial x_j}(x) = y_i(x) \tag{6}$$

Putting the two together means that

$$y_i(x) = 0 \qquad \forall\, x, i \tag{7}$$

Which is equvilant to saying $X = 0$.

part 2

Let f be a smooth, scalar function over a manifold M with $df \equiv 0$. For every point $x \in M$, we can define a coordinate chart (U, φ), with coordinates $x_1, \ldots, x_n$ such that $dx_1, \ldots, dx_n$ is a basis for the cotangent space $T^*_x M$. If in that coordinate chart $f\varphi^{-1} = \phi(x_1, \ldots, x_n)$, where $\phi : \mathbb{R}^n \to \mathbb{R}$, then

$$df_x = \sum_i \frac{\partial \phi}{\partial x_i}(\varphi(x))(dx_i)_x \tag{8}$$

But since $df_x = 0$ for every $x \in M$, we get:

$$\sum_i \frac{\partial \phi}{\partial x_i}(\varphi(x))(dx_i)_x = 0 \tag{9}$$

This is a linear sum of $\{dx_i\}$, which are linearly independent. It vanishes if and only if:

$$\frac{\partial \phi}{\partial x_i}(\varphi(x)) = 0 \qquad \forall\, i \tag{10}$$

So all first order partial derivative vanish in U. Therefore ϕ is constant in U. Which means $f = \phi \circ \varphi$ is locally constant.

Question 4

Let M be a smooth manifold and let $\{(U_i, \varphi_i)\}$ be an atlas over M. For every point p_i in U_i:

$$\varphi_i(p_i) = (x_1(p_i), \ldots, x_n(p_i)) \tag{11}$$

Where $\{x_i\}$ are the coordinates induced by the chart (U_i, φ_i). This induces a set of coordinates for $T^*_{p_i}M$, such that every $v \in T^*_{p_i}M$:

$$v = \sum_j \xi_j (dx_j)_{p_i} \tag{12}$$

The "associated coordinates", or "natural coordinates", are the set of coordinates for the cotangent bundle T^*M that is defined as:

$$\tilde{\varphi}_i(p_i, v) = (x_1(p_i), \ldots, x_n(p_i), \xi_1, \ldots, \xi_n) \in \mathbb{R}^{2n} \tag{13}$$

It is clear to see, that since the atlas $\{(U_i, \varphi_i)\}$ covers all of M, then the set of charts $\{(U_i \times T^*_{U_i}M, \tilde{\varphi}_i)\}$ covers all of T^*M, and also that since for each $i, j \in I$, $U_i \cap U_j$ is an open set, then $U_i \times T^*_{U_i}M \cap U_j \times T^*_{U_j}M$ is also open in $\mathbb{R}^{2n}$, since the product of open sets is open. Finally, we need to show that for each $i, j \in I$, the function $\tilde{\varphi}_i \tilde{\varphi}_j^{-1}$ is smooth in all of it's components. It's easy to see that it is smooth in the first n components since those are the components taken from $\varphi_i \varphi_j^{-1}$. To show that is smooth in the last n components, let $p \in U_i \cap U_j$ and let $v \in T^*_p M$. It can be represented in both coordinates systems dx_j and dy_j:

$$v = \sum_j \xi_j (dx_j)_p \qquad v = \sum_j \xi'_j (dy_j)_p \tag{14}$$

Comparing the two, we get an equality of sums. Applying each sum to $\frac{\partial}{\partial x_k} \in T_pM$ and using the orthogonality of the basis, makes the entire sum on the LHS disappear, except for ξ_k:

$$\xi_k = \sum_j \xi'_j (dy_j)_p \left(\frac{\partial}{\partial x_k}\right)_p = \sum_j \xi'_j (dy_j)_p \left(\frac{\partial}{\partial y_j}\right)_p \left(\frac{\partial y_j}{\partial x_k}\right)_p = \sum_j \xi'_j \left(\frac{\partial y_j}{\partial x_k}\right)_p \tag{15}$$

This is the transition map for the second n components of $\tilde{\varphi}_i \tilde{\varphi}_j^{-1}$, and is clearly smooth. Therefore all transition maps between charts are smooth, and the set $\{(U_i \times T^*_{U_i}M, \tilde{\varphi}_i)\}$ is an atlas over T^*M.

Question 6

Let $f : \mathbb{C} \to \mathbb{C}$ be a smooth map, and let $\mathrm{Re}(f), \mathrm{Im}(f) : \mathbb{C} \to \mathbb{R}$ be the real and imaginary parts of it. We can write it then as:

$$f = \mathrm{Re}(f) + i\,\mathrm{Im}(f) \tag{16}$$

The straightforward coordinate chart $(\mathbb{C}, \varphi : \mathbb{C} \to \mathbb{R})$ such that $\varphi(z) = (\mathrm{Re}(z), \mathrm{Im}(z)) \equiv (x, y)$. This means:

$$\varphi f \varphi^{-1} : \mathbb{R}^2 \to \mathbb{R}^2 \qquad \varphi f \varphi^{-1} = (u(x,y), v(x,y)) \tag{17}$$

Where we define $\mathrm{Re}(f(x+iy)) = u(x,y), \mathrm{Im}(f(x+iy)) = v(x,y)$ for convenience. Now, f is a diffeomorphism if and only if its derivative df is invertible. In the basis induced by φ over T^*M, df can be written as:

$$df = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \tag{18}$$

Which is invertible if and only if

$$\det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \neq 0 \tag{19}$$

On the other hand, since f is smooth, we can write du and dv in the basis of dx and dy:

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy \qquad dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy \tag{20}$$

which makes their wedge product:

$$du \wedge dv = \left(\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy\right) \wedge \left(\frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy\right) = \det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} dx \wedge dy \tag{21}$$

Thus, $du \wedge dv \neq 0$ if and only if the determinant of the Jacobian is non zero, which is true if and only if df is invertible, or in other words that f is a local diffeomorphism.
It is easy to see that this is identical to saying $df \wedge d\bar{f} \neq 0$, since:

$$df \wedge d\bar{f} = (du + idv) \wedge (du - idv) = 2idu \wedge dv \tag{22}$$

Question 7

Let M be a manifold, $p \in M$, and let $X_p : C^\infty(M) \to \mathbb{R}$ be a linear map satisfying the Leibniz rule:

$$X_p(fg) = f(p)X_p(g) + g(p)X_p(f) \qquad \forall\, f, g \in C^\infty(M) \tag{23}$$

Now let $f = gh + c$ where $c \in \mathbb{R}$ is a constant and $f, g \in C^\infty(M)$ vanish at p. Then:

$$X_p(f) = X_p(c) + X_p(gh) = X_p(c) + g(p)X_p(h) + h(p)X_p(g) \tag{24}$$

Because X_p is linear and follows the Leibniz rule, it's easy to see that $X_p(c) = 0$ for any constant $c \in \mathbb{R}$, and because $g(p) = h(p) = 0$, we get:

$$X_p(f) = 0 \tag{25}$$

In other words, when f vanished to the first order in p, we get $X_p(f) = 0$.
So, let (U, φ) be a coordinates chart such that $p \in U$ and let $(x_1, \ldots, x_n)$ be the coordinates induced by it. For a general $f \in C^\infty(M)$, we can write its Taylor expansion using these coordinates:

$$f = f(p) + \sum_i \frac{\partial f}{\partial x_i}(p)x_i + \cdots \tag{26}$$

When acting on f with X_p, we can split it into the zeroth and first order, and all the higher order, but X_p of all the higher orders is zero, since we just saw that X_p vanished if the first order vanishes, giving:

$$X_p(f) = \sum_i \frac{\partial f}{\partial x_i}(p)X_p(x_i) \tag{27}$$

Which is exactly a form of a tangent vector acting on f:

$$v = \sum_i X_p(x_i)\frac{\partial}{\partial x_i} \tag{28}$$

Question 9

Let M be a compact manifold and let V be a smooth vector field on M. Let $(\phi_t)_{t\in\mathbb{R}}$ the transformations associated with V. Since ϕ_t is a local diffeomorphism and from the ODE theorem, for every x there is a neighborhood $x \in U$ and a positive number $\varepsilon > 0$ such that $\phi_\varepsilon(x)$ is well defined. Using compactness, we can choose a uniform ε, for which $\phi_t(x)$ is defined for all $x \in M$.
Once there is a uniform $\varepsilon > 0$, this process can be extended by repeatedly composing ϕ_ε. Therefore, $\phi_t(x)$ is well-defined for all $x \in M$.
Therefore, for every $t \in \mathbb{R}$ and $x \in M$, from the definition of the transformations, $\phi_t(x)$ is a composite function built from the repeated composition of ϕ_ε. Since each of them is a local diffeomorphism, ϕ_{-t} exists globally as well and ϕ_t is a diffeomorphism.

Question 10

Let C be a connected, one-dimensional manifold. Let $\{(U_i, \varphi_i)\}$ be an atlas on C. Using φ_i, every open segment U_i is homeomorphic to an open segment in $\mathbb{R}$. Since C is connected, we can choose a chart (U_0, φ_0) such that for evrey point $p \in C$ exists a finite chain of overlapping charts that will lead from U_0 to the chart containing p. This means we can construct a global smooth map by transporting the local coordinates along the finite number of smooth transition maps of overlapping charts. This gives us a smooth map that spans the entire manifold $\varphi : C \to \mathbb{R}$. Since transition maps and charts are both bijectives and smooth with a smooth inverse, this is a local diffeomorphism.
Let us assume now that C is not compact. From the local diffeomorphism it stems that $\varphi(C)$ is an open interval. Since C is not compact, one can choose the atlas $\{(U_i, \varphi_i)\}$ to have strictly increasing charts. This gives a global order, which makes φ injective, and therefore a diffeomorphism. Therefore $C \cong (0, 1)$.
Let us assume now that C is compact. This means that $\varphi(C)$ is also compact. But the image of a the coordinate

chart φ_0, which is the last map in the composition, is an open set. There are no open sets on $\mathbb{R}$ that are compact, and so φ cannot be injective. In other words, there exist points $p \neq q \in C$ such that $\varphi(p) = \varphi(q)$. Since C is connected, there is a continous path l from p to q. The image of that path $\varphi(l) \in \mathbb{R}$ is a covering map of S^1. Therefore $C \cong S^1$.